SHALEEN JAYARAJ https://www.linkedin.com/in/shaleen-jayaraj/

CAE Engineering Intern Candidate

Graduate engineering student with hands-on experience in CAE modeling, vehicle simulation, suspension analysis, and full-vehicle performance analysis. Skilled in applying engineering principles to real-world automotive projects, with strong expertise in ADAMS, SolidWorks, and vehicle dynamics modeling. Motivated to contribute to durability, CAE, and vehicle performance teams through data-driven engineering, continuous learning, and effective collaboration.

Greenville, South Carolina | shaleej@clemson.edu | +1 864-451-8757

Education

Program	Study Level	University/Institute	CGPA	Year
Automotive Engineering	Masters	Clemson University – CUICAR	3.42/4	2025-27
Mechanical Engineering	Bachelors (Honors)	Rajiv Gandhi Institute of Technology, APJAKTU	8.32/10	2018-22

Courses: Dynamic Performance of Vehicles, Control Systems Engineering, Vehicle Testing & Characterization (Modal Analysis, Chassis Dyno, 4-Post Rig, Vehicle Testrack: CRC, Moose, Braking, Swept-Sine), Automotive Systems Overview

Project Work(AUE Clemson):

Ride and Handling Analysis of 2-Axle and 3-Axle Trucks Using Bicycle and Ride Models.

- Used PSD and RMS acceleration metrics to evaluate ride comfort of 2-axle and 3-axle trucks.
- Built bicycle models and performed transient handling analysis, comparing understeer gradient, yaw response, and stability characteristics.

Vehicle Dynamics Data Analysis & Tire Model–Based Simulation

- Analyzed vehicle test data to extract tire parameters using binning and curve fitting.
- Built a lateral tire force model and integrated it into a bicycle vehicle dynamics model.
- Simulated yaw and lateral response and validated results against experimental data.
- Performed data acquisition and post-processing using MATLAB and Siemens Testlab to extract frequency response, ride comfort metrics, and dynamic characteristics.

Professional Experience

Vehicle Dynamics Engineer - Applus+ IDIADA(TATA Motors ODC) — Pune, India | 10th Aug'22 – 25th Apr'25

- Led transient and steady-state vehicle dynamics analyses (cornering, braking, step steer, ride) to evaluate and improve handling and stability for passenger vehicles.
- Developed full-vehicle multibody models in ADAMS/View using Template Builder, Event Builder, and Road Builder, including flex-body integration for suspension and chassis components.
- Performed vehicle settling procedures in simulation environments to establish accurate static equilibrium conditions before dynamic events.
- Utilized understeer budgeting sheets to quantify steering balance and guide suspension tuning decisions for improved handling performance.
- Executed Kinematics & Compliance (K&C) studies to quantify bump steer, roll steer, camber compliance, and wheel rates; proposed geometry changes and bushing changes to achieve handling predictability and steering response.
- Applied Design of Experiments (DOE) in ADAMS/Insight to optimize suspension hardpoints and meet ride/handling targets across multiple weight and loading conditions.
- Diagnosed steering wobble and rack-load issues; optimized hydrobush stiffness and steering link geometry, reducing rack load and achieving $25 USD/vehicle cost savings.
- Released suspension tuning data sheets for various curb-to-GVW load conditions, ensuring compliance with vehicle attribute targets and manufacturing tolerances.
- Performed key geometric and durability checks: wheel/ARB ball-joint envelopes, driveshaft pull-out (DSPO), service loads, Virtual RLDA, and ground-clearance audits.
- Developed automation tools to streamline CAE workflows and collaborated cross-functionally with chassis, steering, and CAE teams to validate results and deliver correlation reports with test data.

Intern | Vehicle Dynamics Scholar, Capabl — 1st June'20 – 31st August20

- Gained comprehensive knowledge in Vehicle Dynamics during the internship, focusing on designing an all-terrain vehicle (ATV).
- Designed a steering system and verified calculations graphically using SolidWorks.
- Analyzed the effects of steering, bump, and roll motions on suspension parameters like caster, camber, and toe using Lotus Shark software.
- Modeled and conducted analysis for multiple components of an ATV to optimize performance and reliability.

Academic works & Publication (Undergrad)

Academic major Project: Self Tuning setup enhancement for an autonomous car using Machine Learning. This study focused on finding the best tuning setup for a target vehicle using Machine Learning.

Academic Seminar: A detailed study on design, fabrication, analysis and testing of the Anti -Roll bar system for formula student cars.

Publication: *Design and Modelling of Height Adjustable Grab Holder* - International Research Journal of Engineering and Technology

Leadership Experience

Suspension Head Team Motorage BAJA SAE INDIA

- Led the design of suspension geometry using SolidWorks, followed by kinematic analysis using Lotus Shark.
- Conducted structural analysis of suspension components using ANSYS to ensure durability and performance.
- Modeled the vehicle in IPG Carmaker and performed virtual simulations to optimize vehicle dynamics and handling performance.
- Coordinated with other subsystem teams to ensure seamless integration of the suspension system with chassis and drivetrain.

SEC Member at SAE RIT.

- I worked as a student executive member for the college chapter of SAE RIT.
- Collaborated with SAE societies across the state to organize national-level automotive events.

Certifications & Skills

ADM740 - Vehicle Modeling and Simulation using Adams Car - MSC Software

ADM704B - Automating Tasks using Adams/View Scripting, Macros, and GUI Customization - MSC Software

ADM705 - Python Scripting in Adams

Simulation : - Adams Car, Adams View, IPG Carmaker

Programming : - MATLAB, Python, Adams CMD

CAD Modelling : - SolidWorks, Siemens NX

Online Self Learned Courses: Fundamental of Vehicle Dynamic - NPTEL IIT Madras | Machine Dynamic using MATLAB - RWTH Aachen | Python for Everybody - Coursera - University of Michigan | Introduction to Programming with MATLAB - Coursera - Vanderbilt University | Machine Design - Coursera - Georgia Institute of Technology